

12.421/12.621 Lecture 10: Interaction of EM Radiation with Atmospheres

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Topics covered

- Review of Earth's atmosphere structure
- Brief discussion of temperature lapse rate and pressure and density profiles
- Absorption and emission by gases in the atmosphere

1 Interaction of EM radiation and atmospheres

- Virtually all remote sensing applications need to contend with atmosphere, whether the atmosphere is signal or noise.
- As we saw, the transparency of the atmosphere varies with EM frequency from virtually opaque to virtually transparent in a way that may appear at first glance to be quasi-random.
- To understand how the atmosphere can be observed with remote sensing and can influence terrestrial observations, we start with basic definitions of the atmosphere.

2 Standard atmosphere of Earth

- Earth's atmosphere is made up of 5 primary layers. Layer boundaries are defined by changes in thermal lapse rate. From top to bottom, the layers are
 - Exosphere
 - Altitude range ~ 700 km (AMSL) – $\sim 10^4$ km
 - Extremely low density
 - bottom is defined by “exobase” (also known as thermopause) $\sim$ orbital altitude of many Earth-observing satellites
 - low density + high altitude $\rightarrow$ negligible for most RS applications
 - Thermosphere
 - Altitude range ~ 80 km – ~ 700 km
 - low density (many satellites and ISS orbit here)
 - temperature increases with elevation (temperature inversion) from ≈ 190 K to ≈ 330 K (in International Standard Atmosphere; 330 K is reached at 120 km altitude, where the standard profile of Ulaby and Long (2014), Fig. 8-3, ends; higher up, temperatures reach 500–2000 K, depending on solar activity)
 - top is defined by exobase (a.k.a. thermopause)

- bottom defined by mesopause
- Importantly for RS applications, ionosphere is located at the bottom of thermosphere
- ionosphere is the ionized layer of Earth's atmosphere; plays a key role in EM wave propagation from satellites and radio propagation on Earth
- ionosphere is a layer of free electrons; electrons are stripped from atoms and molecules of the atmosphere by ultraviolet and higher frequency radiation from the Sun; electrons remain free for some time due to low density of thermosphere; eventually, electron will be recaptured by a positive ion (process called recombination); low density enables rate of release of electrons $\sim$ rate of recombination with positive ions $\Rightarrow$ persistent plasma (ionosphere) with temporal variability
- ionosphere is dispersive at microwave frequencies, enables detailed studies using dual-frequency GPS
- Strongest influence on RS is at very high and low (microwave and lower) frequencies
- Mesosphere
 - Altitude range ~ 50 km – 80 km
 - Temperature decreases with elevation (+ lapse rate) from ≈ 270 K at the base down to ≈ 190 K at the mesopause
 - top defined by mesopause
 - base defined by stratopause
 - Can allow for formation of ice-crystal clouds, called noctilucent clouds
- Stratosphere
 - Altitude range ~ 12 km (tropopause) – ~ 50 km (stratopause)
 - Temperature increases with altitude (i.e., – lapse rate or temperature inversion) from ≈ 213 K at the tropopause to ≈ 270 K at the stratopause (in the standard atmosphere, the temperature is constant from the tropopause to ~ 20 km and increases above)
 - Temperature inversion caused by absorption of ultraviolet radiation by the ozone layer
 - Very dry, though some water vapor produced in situ by photochemical oxidation of methane
 - No weather (because no turbulence), but polar stratospheric or nacreous clouds can form in winter and at high latitudes
- Troposphere ('tropo' is Greek for 'turn')
 - Altitude range from Earth's surface to ~ 12 km (tropopause)
 - Upper altitude varies as a function of latitude and other factors, such as weather
 - In general, temperature decreases with altitude due to heating from Earth's surface; temperature inversions can and do occur
 - Accounts for vast majority of total mass of atmosphere $\Rightarrow$ most dense part of atmosphere
 - Holds virtually all atmospheric moisture $\Rightarrow$ can have major impact on most forms of RS
 - All weather and virtually all clouds are found in troposphere $\Rightarrow$ object of many RS observations and noise source for others

★ To a good approximation in most RS applications, we can consider the atmosphere to be the ionosphere and the troposphere.

3 Properties of the troposphere

3.1 Temperature

- Common temperature profiles are defined by International, U.S., or other Standard Atmospheres.
- In the U.S. and International Standard Atmospheres, temperature is linear with elevation in the troposphere.
- The slope is known as the “temperature lapse rate”, a function of moisture content.
- For a dry atmosphere, 1st Law of Thermodynamics for adiabatic process given as

$$\rho_{\text{air}} c_p dT = dP, \quad (1)$$

$T \equiv$ temperature,

$P \equiv$ pressure,

$\rho_{\text{air}} \equiv$ mass density of dry air,

$c_p \equiv$ specific heat (constant pressure).

Assume hydrostatic

$$\Rightarrow dP = -\rho_{\text{air}} g dz, \quad g \equiv \text{gravitational acceleration.} \quad (2)$$

Dry adiabatic lapse rate is

$$a_d = -\frac{dT}{dz} = \frac{g}{c_p} \approx \frac{9.81 \text{ m/s}^2}{1000 \text{ J/(kg K)}} \approx 9.8 \frac{\text{K}}{\text{km}} \quad (\text{a constant}). \quad (3)$$

- Moisture in the atmosphere complicates the lapse rate; must account for condensation (involves release of latent heat)

$$\begin{aligned} \Rightarrow a_{\text{moist}} &\equiv \text{moist adiabatic lapse rate} \\ &= a_{\text{moist}}(T, P) \quad \text{most strongly dependent on } T. \end{aligned}$$

- Standard atmosphere does not explicitly account for moisture, but sets constant (within troposphere) temperature lapse rate

$$\boxed{a \equiv 6.5 \frac{\text{K}}{\text{km}}} \quad (4)$$

3.2 Pressure

- Can be shown that pressure varies with altitude z as

$$P = P_0 \left(1 - \frac{az}{T_0} \right)^{\frac{gM}{aR_{\text{gas}}}}, \quad (5)$$

$a \equiv$ temperature lapse rate,

$T_0 \equiv$ sea-level standard temperature,

$P_0 \equiv$ sea-level pressure,

$M \equiv$ molar mass of dry air,
 $R_{\text{gas}} \equiv$ universal gas constant.

- Assuming $a = a_d = g/c_p \equiv$ dry adiabatic lapse rate

$$P = P_0 \left(1 - \frac{gz}{c_p T_0} \right)^{\frac{c_p M}{R_{\text{gas}}}}. \quad (6)$$

- Given $c_p \sim 10^3 \text{ J/(kg K)}$, $M \sim 2.9 \times 10^{-2} \text{ kg/mol}$, and $R_{\text{gas}} = 8.314 \text{ J/(mol K)}$, binomial approx gives

$$P \approx P_0 e^{-z/H_p}, \quad H_p = \frac{R_{\text{gas}} T_0}{gM} \approx 8.4 \text{ km} \equiv \text{pressure scale height of atmosphere} \quad (7)$$

3.3 Density

- The identical argument can be made for density of dry air as for pressure

$$\Rightarrow \rho_{\text{air}} = \rho_{\text{air},0} e^{-z/H_p} \quad (8)$$

where $H_p \approx H_\rho$ (equal for an isothermal atmosphere; for the standard atmosphere H_ρ is somewhat larger) and $\rho_{\text{air},0} \equiv$ density of dry air at sea level.

- Water vapor density varies in space and time, standard atmosphere gives

$$\rho_{\text{vap}} = \rho_{\text{vap},0} e^{-z/H_{\text{vap}}}, \quad (9)$$

where $\rho_{\text{vap},0} \equiv 7.72 \times 10^{-3} \text{ kg/m}^3$ and $H_{\text{vap}} \approx 2 \text{ to } 2.5 \text{ km}$.

4 Absorption and emission by gases

- Brief (very high-level) overview of the fundamental concepts of absorption and emission by atmospheric gases.
- Total internal energy $\mathcal{E}$ of isolated molecule consists of 3 energy states

$$\mathcal{E} = \mathcal{E}_e + \mathcal{E}_v + \mathcal{E}_r, \quad (10)$$

$\mathcal{E}_e \equiv$ electronic energy $\rightarrow$ electron energy levels in individual atoms,
 $\mathcal{E}_v \equiv$ vibrational energy $\rightarrow$ change of position between two atoms,
 $\mathcal{E}_r \equiv$ rotational energy $\rightarrow$ rotation of molecule.

- Energy states are quantized $\Rightarrow$ take on discrete values specified by one or more quantum numbers.
- Let us consider each energy state at a very high level
 - For illustrative purposes, let us consider diatomic molecules
 - A note on bonding:
 - Ionic bonding – electrostatic attraction between atoms – one atom loses an outer electron (+ charge) and the other gains electron (– charge). E.g., sodium chloride (sodium loses electron) (Figure 1a).

- Covalent bonding – atoms share outer electrons. E.g., Figure 1b.
- With either type of bond, we may expect a gradient in electric charge $\Rightarrow$ polarization, or electric dipole $\Rightarrow$ applied electric field can cause rotation (but not for molecules made of two identical atoms, such as N_2 and O_2 , which have no permanent electric dipole moment; O_2 has a magnetic dipole moment instead).

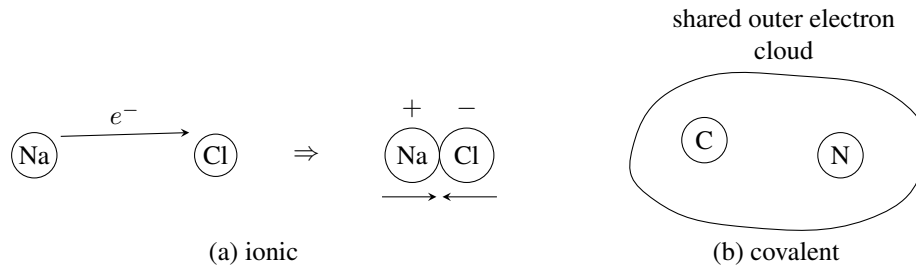


Figure 1: (a) Ionic bonding: sodium gives an outer electron to chlorine, and the resulting ions attract each other. (b) Covalent bonding: the atoms share their outer electrons.

- Different energy states
 - Rotational energy
 - $\hookrightarrow$ Our diatomic molecule can rotate “end over end” but can not rotate about axis connecting atoms (Figure 2).

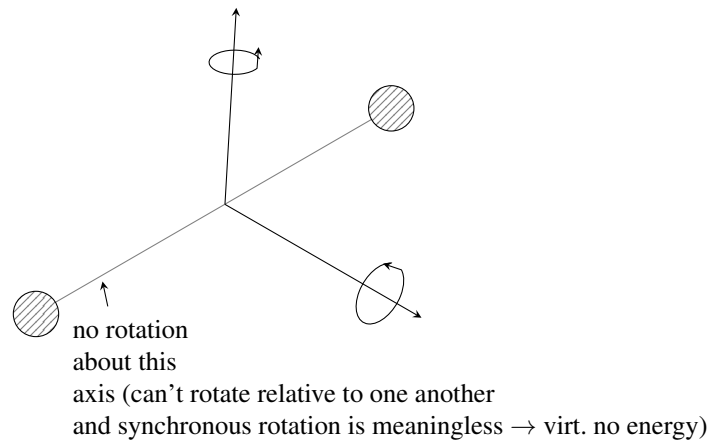


Figure 2: A diatomic molecule rotates “end over end” about the two axes perpendicular to the line joining the atoms, but not about that line.

- $\hookrightarrow$ laws of quantum mechanics require rotations only at discrete rates or frequencies (not continuous as in classical mechanics)
- $\hookrightarrow$ But like classical mechanics, higher rotation rates imply higher energy
- $\hookrightarrow$ discrete rotation rates and rates $\leftrightarrow$ energy $\Rightarrow$ described by “rotation levels” $\rightarrow$ energy state $\mathcal{E}_r$
- $\hookrightarrow$ Key feature is that rotational transitions can occur by absorbing or emitting photons, * but can only ‘jump’ from one rotation level to another and only one at a time *

- ↪ To move from one rotation level to another, molecule must absorb a photon with frequency equal to that of the next rotation level (frequency) $\Rightarrow$ to spin faster, molecules must absorb increasingly higher frequency EM radiation
- ↪ (conversely,) to slow down, a molecule must emit EM radiation with frequency equal to the rotation level
- ↪ rotation levels are not equally spaced in energy ($\mathcal{E}_r \propto J(J+1)$), but the energy differences between adjacent levels increase in equal steps ($\propto J+1$), and hence so do the frequencies of the allowed one-level transitions; therefore absorption lines are equally spaced in frequency but not wavelength (Figure 3).

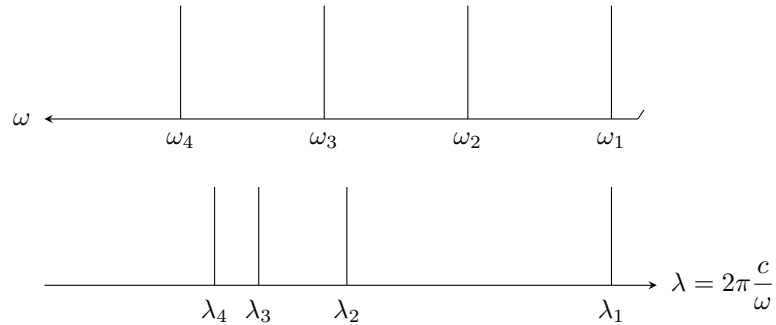


Figure 3: Pure rotational absorption lines are equally spaced in frequency (top) but not in wavelength (bottom). The line positions are drawn with $\omega_2 = 2\omega_1$, $\omega_3 = 3\omega_1$, $\omega_4 = 4\omega_1$, so $\lambda_2 = \lambda_1/2$, $\lambda_3 = \lambda_1/3$, $\lambda_4 = \lambda_1/4$.

- ↪ $\mathcal{E}_r = \omega_{\text{rot}} \hbar \sqrt{J(J+1)}/2$, where $\hbar \equiv$ reduced Planck constant ($\hbar = h/2\pi$), $J \equiv$ nonnegative integer called the rotational quantum number, and $\omega_{\text{rot}} \equiv$ angular velocity of the rotation.
- Vibrational energy $\mathcal{E}_v$
 - ↪ dipoles can be ‘smushed’ together along the line joining the atoms
 - ↪ basic process (Figure 4)

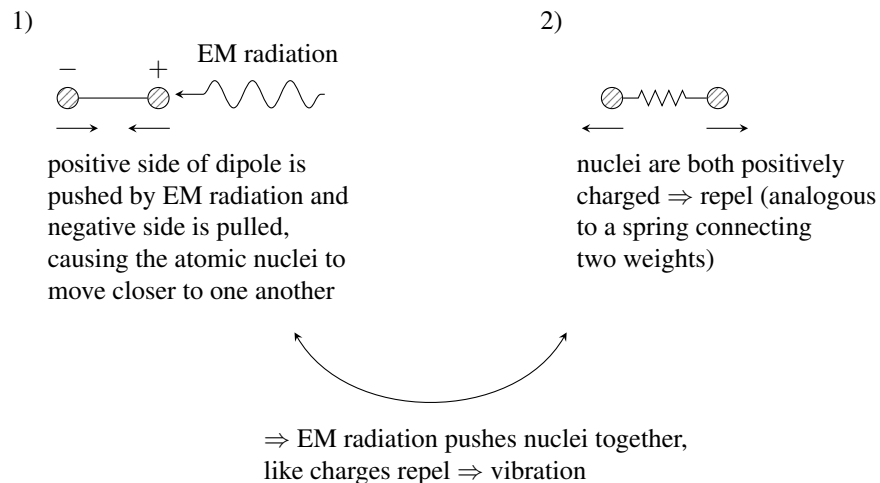


Figure 4: Basic process of molecular vibration driven by EM radiation.

- ↪ frequency of vibrations are quantized → vibration levels
- ↪ When a molecule absorbs EM radiation of the frequency associated with a higher vibration level, a vibration transition occurs
- ↪ * However * vibration transitions are always accompanied by a rotation transition → as we saw in the previous section, rotation transitions involve jumps of one and only one level (can be up or down) (Figure 5)

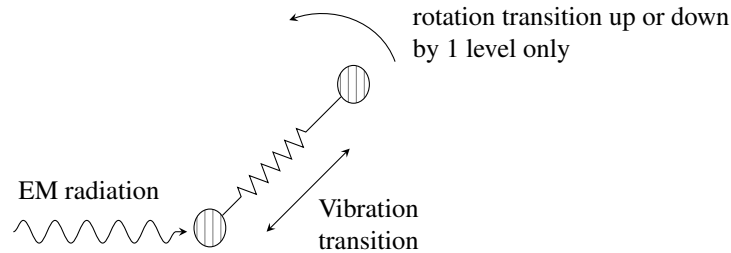


Figure 5: A vibration transition driven by EM radiation, accompanied by a rotation transition of one level.

- ↪ Since each vibration level can be associated with several rotation (sub-)levels, there are many possible vibration transitions within a population
- ↪ All vibration transitions require slightly different energy levels, results in a series of closely spaced absorption lines that converge at short wavelengths (i.e., higher frequencies) ⇒ absorption bands
- ↪ Key point is that a range of higher frequency EM can be absorbed by molecules in the atmosphere → visible red to thermal infrared portion of EM spectrum
- ↪ $\mathcal{E}_v = (n_{\text{vib}} + \frac{1}{2}) \hbar \omega_{\text{vib}}$, where n_{vib} is an integer called the vibrational quantum number (and ω_{vib} is the natural angular frequency of the vibration).

Variables

Units are SI. Values are given for universal constants; “typical” values are representative, not definitive.

Symbol	Description	Units	Value
<i>Constants</i>			
c	speed of light in a vacuum	m/s	2.998×10^8
g	gravitational acceleration	m/s ²	9.81
R_{gas}	universal gas constant	J mol ⁻¹ K ⁻¹	8.314
h	Planck constant	J s	6.626×10^{-34}
$\hbar$	reduced Planck constant, $h/2\pi$	J s	1.055×10^{-34}
<i>Atmosphere</i>			
z	altitude (height above sea level)	m	–
T	temperature	K	–
T_0	sea-level standard temperature	K	288.15
P	pressure	Pa	–
P_0	sea-level pressure	Pa	1.013×10^5 (1013.25 mbar)
ρ_{air}	mass density of dry air	kg/m ³	–
$\rho_{\text{air},0}$	density of dry air at sea level	kg/m ³	1.225
c_p	specific heat of dry air at constant pressure	J kg ⁻¹ K ⁻¹	$\sim 10^3$ (1004)
M	molar mass of dry air	kg/mol	2.9×10^{-2}
a	temperature lapse rate, $-dT/dz$	K/m	6.5 K/km (standard atmosphere)
a_d	dry adiabatic lapse rate, g/c_p	K/m	≈ 9.8 K/km
a_{moist}	moist adiabatic lapse rate	K/m	–
H_p	pressure scale height, $R_{\text{gas}}T_0/(gM)$	m	≈ 8.4 km
H_ρ	density scale height	m	$\approx H_p$
ρ_{vap}	water-vapor density	kg/m ³	–
$\rho_{\text{vap},0}$	water-vapor density at sea level (standard atmosphere)	kg/m ³	7.72×10^{-3}
H_{vap}	water-vapor scale height	m	2–2.5 km
<i>Molecular energy states</i>			
$\mathcal{E}$	total internal energy of a molecule	J	$\mathcal{E}_e + \mathcal{E}_v + \mathcal{E}_r$
$\mathcal{E}_e, \mathcal{E}_v, \mathcal{E}_r$	electronic, vibrational and rotational energy	J	–
e^-	electron	–	–
J	rotational quantum number	dimensionless	0, 1, 2, ...
n_{vib}	vibrational quantum number	dimensionless	0, 1, 2, ...
ω	angular frequency of the EM radiation (Figure 3)	rad/s	–
ω_{rot}	angular velocity of a molecule’s rotation	rad/s	–
ω_{vib}	natural angular frequency of a molecular vibration	rad/s	–
$\omega_1, \dots, \omega_4$	angular frequencies of rotational absorption lines 1–4	rad/s	$\omega_2 = 2\omega_1$, etc. (rigid rotor)
$\lambda; \lambda_1, \dots, \lambda_4$	wavelength; wavelengths of lines 1–4	m	$2\pi c/\omega$